Baibhav Pathak

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Summary

Senior Software Engineer specializing in backend systems, automation platforms and API development with Python, C++, Node.js and cloud technologies. Experienced in building scalable services, CI/CD workflows and data pipelines with cross-functional teams, delivering reliable, maintainable software.

Experience

Zoppler Systems · *Senior Software Engineer · Bangalore, India* **Oct 2026 – Present**
C++, Python

IpserLabs · *Software Engineer · Boston, USA* **Sep 2025 – Jul 2026**

- Designed and enhanced backend services for control-plane workflows in Python and Go, supporting **1K+ daily active users**.
- Built secure REST APIs with **JWT authentication and role-based access control** for registration, resource configuration and interaction workflows.
- Automated CI/CD with GitHub workflows and Python, speeding up deployments by **20%** across staging and production.

PPD, part of Thermo Fisher Scientific · *Software Engineer, Backend & Automation · Bangalore, India* **Jan 2021 – Aug 2023**

- Built and deployed **50+ backend automation solutions** in Python, PowerShell and shell, serving **10,000+ users** globally, end to end from design to maintenance.
- Developed REST APIs and data pipelines on AWS and Azure with QA, operations and 200+ end users, improving data accuracy by **35%** and operational efficiency by **30%**.
- Led Azure SQL data extraction via Microsoft Graph API, ran Jenkins CI/CD, kept a **95% resolution rate** at third-level support and mentored junior engineers.

Education

Northeastern University · *Master of Science in Internet of Things* **Aug 2023 – May 2025**
Visvesvaraya Technological University · *Bachelor of Technology in Computer Science* **Aug 2016 – Sep 2020**

Projects

baibhavOS, portfolio as a browser OS | *Next.js, TypeScript, Zustand, Vitest* **Oct 2026 – Present**

- Window manager with 12 apps, a URL per app and a phone layout on one store; live MBTA and GitHub data through cached API routes; **90+ tests** in CI.

Real-Time ASL Gesture Interpretation Engine | *Python, Flask, TensorFlow, MediaPipe* **Mar 2024 – Apr 2024**

- Flask REST API serving a Random Forest + TensorFlow ensemble: **2.7 ms median inference**, **100% held-out accuracy** across 27 classes.
- Scaled to **3.5× throughput** (443 → 1,569 req/s) and cut p95 latency by **57%** with micro-batching, gunicorn workers and an LRU cache.

Boston Subway Network Optimization & Visualization | *Python, FastAPI, BeautifulSoup* **Feb 2024 – Mar 2024, Oct 2026**

- FastAPI route engine over 125 MBTA stations: Dijkstra across all **7,750 station pairs**, **9× faster** and **62% less memory** than the matrix version.
- Live-alert pipeline (MBTA V3 API + scraper) reroutes trips around closures; LRU route cache with a **77% hit rate**.

Vision Transformer SoC System | *C++, SystemC, FPGA* **Nov 2025 – Dec 2025**

- Edge image classifier with parallel patch embeddings and fixed-point arithmetic, cutting inference latency by **30%**; FPGA feasibility studies.

Cellular-Connected Autonomous Drone | *C++, Python, Arduino, TCP/UDP* **Nov 2019 – Mar 2020**

- Quadcopter with cellular tower routing (**+30% range**), IMU/GPS/barometer/camera over SPI, I2C and UART, and live TCP/UDP telemetry.

Skills

Languages: Python, C++, Go, TypeScript, JavaScript, C, SQL, PowerShell, Shell, SystemC, HTML5, CSS3
Backend & Cloud: FastAPI, Flask, Node.js, Express, REST, JWT, WebSockets, AWS (EC2, S3, Lambda), Azure, Docker, Jenkins, GitHub Actions
Data & ML: PostgreSQL, MongoDB, MySQL, Azure SQL, TensorFlow, scikit-learn, OpenCV, MediaPipe, NumPy
Embedded & IoT: Arduino, Raspberry Pi, UART, I2C, SPI, TCP/IP, UDP, MQTT